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Ex No 8 : Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim

To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

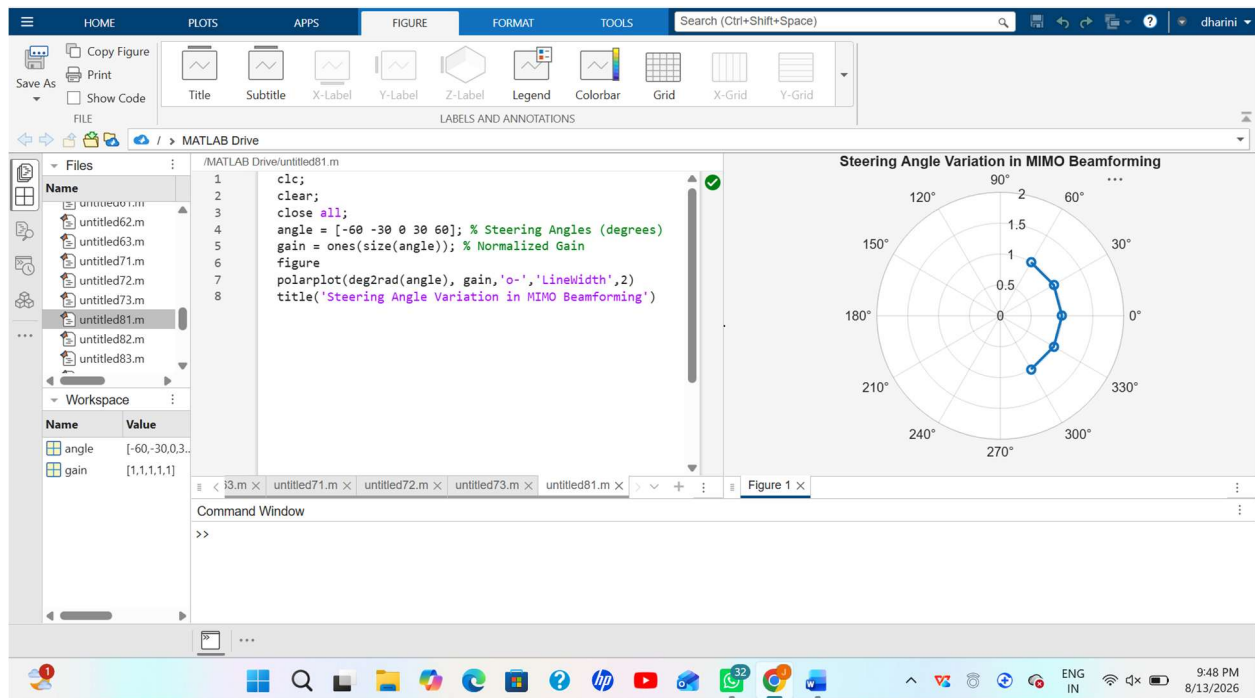
Objectives

1. To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.
2. To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.
3. To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

Procedure:

```
clc;  
clear;  
close all;  
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)  
gain = ones(size(angle)); % Normalized Gain  
figure  
polarplot(deg2rad(angle), gain,'o-','LineWidth',2)  
title('Steering Angle Variation in MIMO Beamforming')
```

Output:



Objective 2:

```
clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = [9 12 15 12 9]; % Array Gain (dB)

figure

bar(angle, gain)

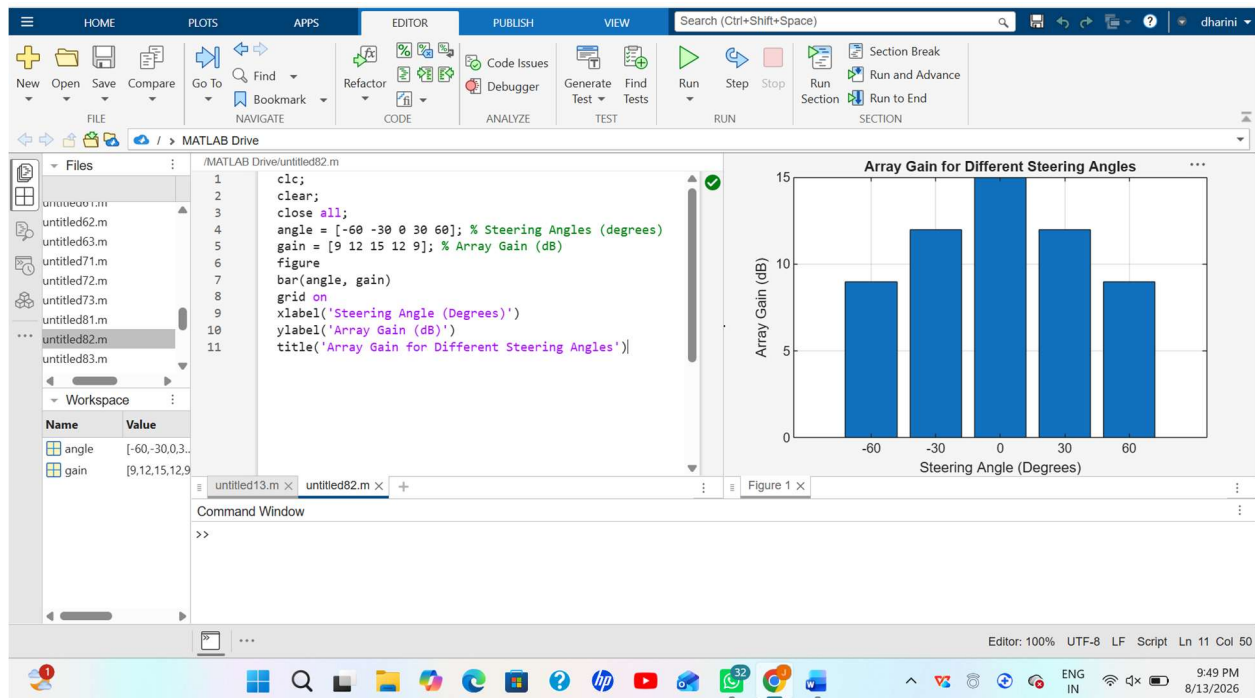
grid on

xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')

title('Array Gain for Different Steering Angles')
```

Output:



Objective 3:

`clc;`

`clear;`

`close all;`

`desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)`

`actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)`

`error = abs(desired - actual); % Beam Direction Error`

`figure`

`plot(desired, error, 'o-', 'LineWidth', 2)`

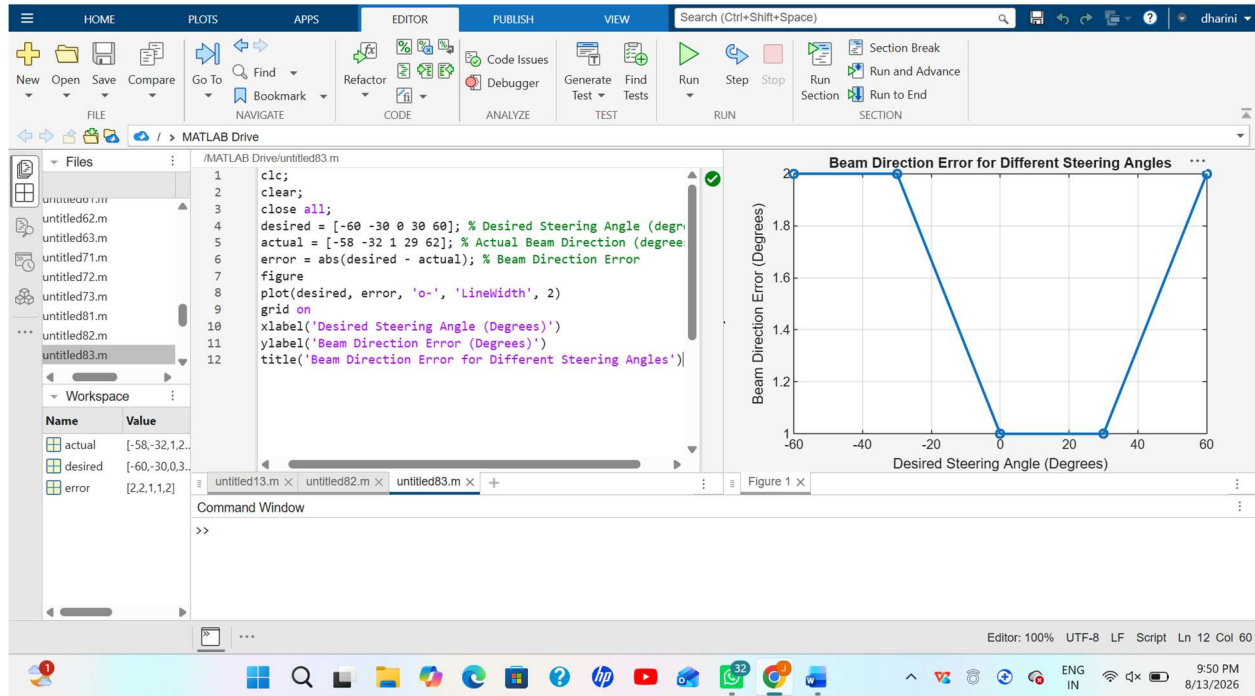
`grid on`

`xlabel('Desired Steering Angle (Degrees)')`

`ylabel('Beam Direction Error (Degrees)')`

`title('Beam Direction Error for Different Steering Angles')`

Output:



Result

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error.